

Banking System Apl

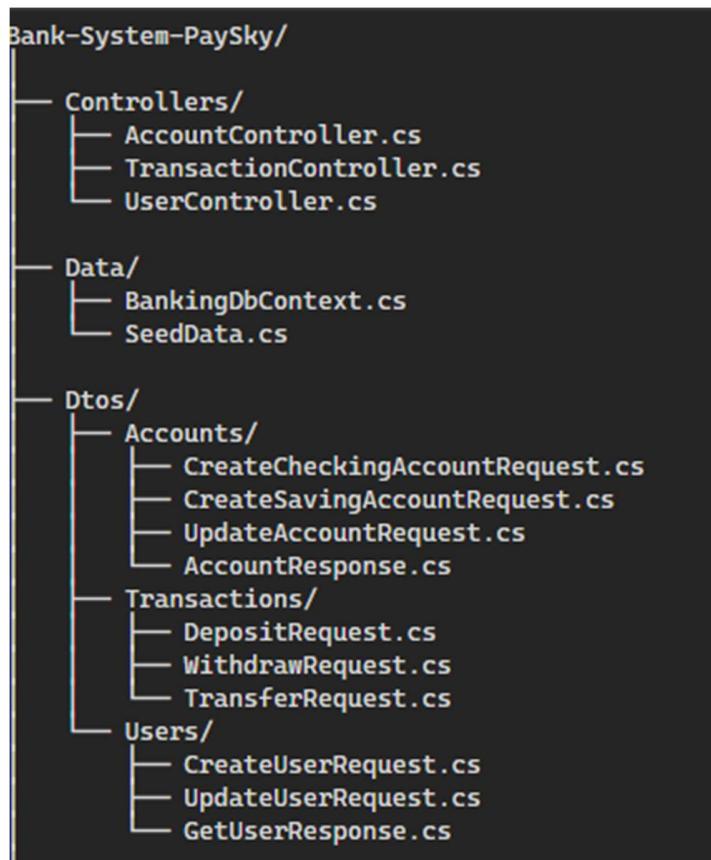
I would like to take this opportunity to express my heartfelt thanks to PaySky for providing me with the chance to work on this project. I truly appreciate the opportunity to contribute and learn from such an esteemed company.

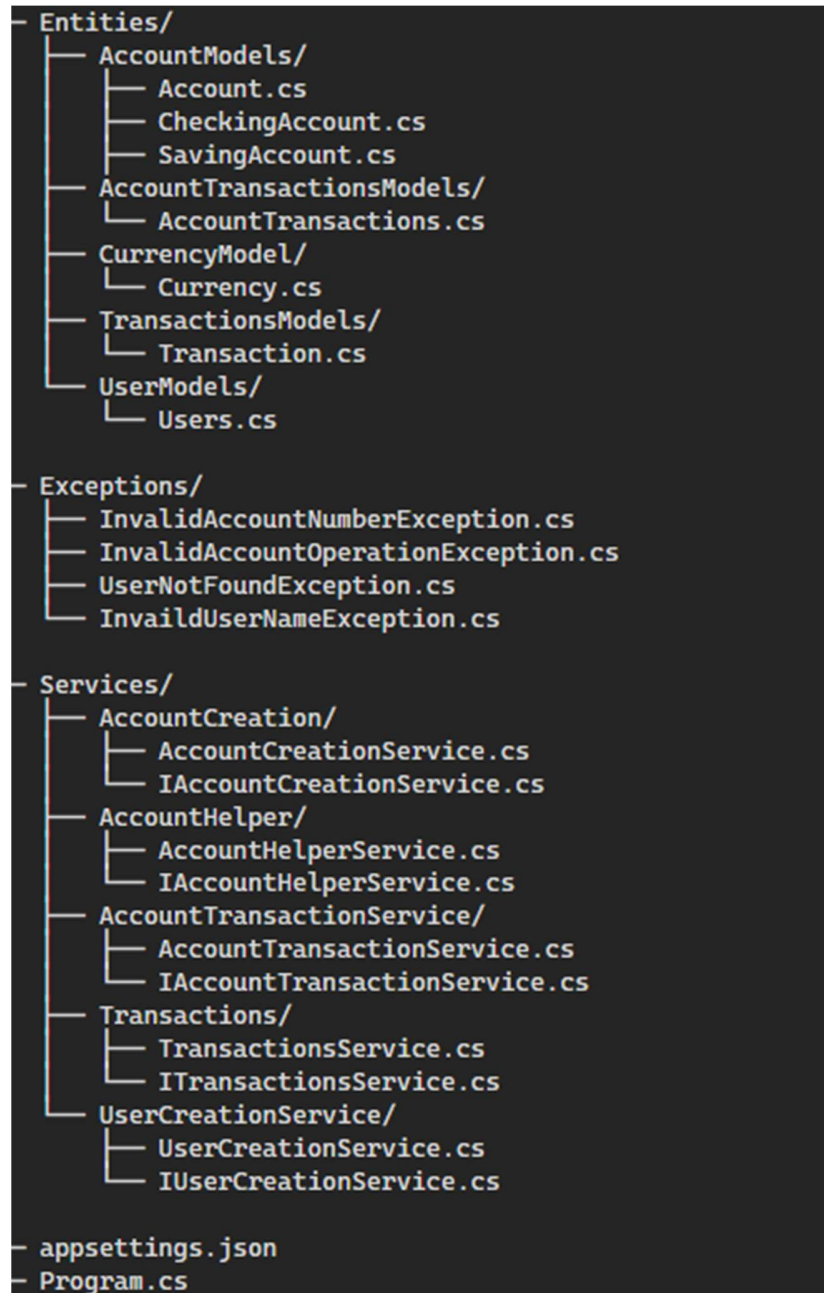
After carefully reviewing the project requirements and completing all the necessary tasks for the Banking System API

This ALL about what I did in this project this is a solution contain two applications:

- 1- Bank-System-PaySky → this the main Apl project with .NET 8 Web API
Configure Program.cs using the Minimal Hosting Model without requiring Startup.cs.
- 2- UnitTest → tasting application work with NUnit

Bank-System-PaySky





Controllers

- **AccountController.cs**: Handles HTTP requests related to account operations such as creating, updating, and deleting accounts.
- **TransactionController.cs**: Manages HTTP requests for transaction operations like deposits, withdrawals, and transfers.

- **UserController.cs**: Manages HTTP requests for user operations such as creating, updating, and deleting users.

Data

- **BankingDbContext.cs**: Defines the Entity Framework Core database context, which manages the database connections and entity sets.
- **SeedData.cs**: Contains methods to seed the database with initial data.

Dtos (Data Transfer Objects)

- **Accounts**: Contains DTOs related to account operations.
 - **CreateCheckingAccountRequest.cs**: DTO for creating a checking account.
 - **CreateSavingAccountRequest.cs**: DTO for creating a saving account.
 - **UpdateAccountRequest.cs**: DTO for updating an account.
 - **AccountResponse.cs**: DTO for returning account details.
- **Transactions**: Contains DTOs related to transaction operations.
 - **DepositRequest.cs**: DTO for deposit requests.
 - **WithdrawRequest.cs**: DTO for withdrawal requests.
 - **TransferRequest.cs**: DTO for transfer requests.
- **Users**: Contains DTOs related to user operations.
 - **CreateUserRequest.cs**: DTO for creating a user.
 - **UpdateUserRequest.cs**: DTO for updating a user.
 - **GetUserResponse.cs**: DTO for returning user details.

Entities

- **AccountModels**: Contains entity classes related to accounts.
 - **Account.cs**: Base class for account entities.
 - **CheckingAccount.cs**: Entity class for checking accounts.
 - **SavingAccount.cs**: Entity class for saving accounts.
- **AccountTransactionsModels**: Contains entity classes related to account transactions.
 - **AccountTransactions.cs**: Entity class for account transactions.
- **CurrencyModel**: Contains entity classes related to currencies.
 - **Currency.cs**: Entity class for currencies.
- **TransactionsModels**: Contains entity classes related to transactions.
 - **Transaction.cs**: Entity class for transactions.

- **UserModels:** Contains entity classes related to users.
 - **Users.cs:** Entity class for users.

Exceptions

- **InvalidAccountNumberException.cs:** Custom exception for invalid account numbers.
- **InvalidAccountOperationException.cs:** Custom exception for invalid account operations.
- **UserNotFoundException.cs:** Custom exception for user not found.
- **InvalidUserNameException.cs:** Custom exception for invalid user names.

Services

- **AccountCreation:** Contains services related to account creation.
 - **AccountCreationService.cs:** Service class for account creation operations.
 - **IAccountCreationService.cs:** Interface for account creation service.
- **AccountHelper:** Contains helper services for account operations.
 - **AccountHelperService.cs:** Helper service class for account operations.
 - **IAccountHelperService.cs:** Interface for account helper service.
- **AccountTransactionService:** Contains services related to account transactions.
 - **AccountTransactionService.cs:** Service class for account transaction operations.
 - **IAccountTransactionService.cs:** Interface for account transaction service.
- **Transactions:** Contains services related to transactions.
 - **TransactionsService.cs:** Service class for transaction operations.
 - **ITransactionsService.cs:** Interface for transaction service.

- **UserCreationService**: Contains services related to user creation.
 - **UserCreationService.cs**: Service class for user creation operations.
 - **IUserCreationService.cs**: Interface for user creation service.

Configuration Files

· **appsettings.json**: Configuration file for application settings, including connection strings and logging settings.

Entry Point

· **Program.cs**: The main entry point for the application.